Sufficiency, Convexity, and the Rao–Blackwell Theorem

Stats 220B

Recap from Lecture 1

We set up the statistical decision problem: Θ, $\mathcal{A}$, $L(\theta, a)$, data $X \sim P_\theta$, prior $\pi(\theta)$.

We defined decision rules $\delta : \mathcal{X} \to \mathcal{A}$, and introduced two ways to evaluate them:

- **Frequentist risk:** $R(\theta, \delta) = E_{P_\theta}[L(\theta, \delta(X))]$. A function of θ, not a number.
- **Bayesian expected loss / Bayes risk:** $\rho(\pi, a) = E_\pi[L(\theta, a)]$ and $r(\pi, \delta) = E_\pi[R(\theta, \delta)]$.

We defined admissibility, saw that risk functions cross, and introduced three selection principles: Bayes, minimax, invariance.

We also introduced randomized decision rules $\delta^*(x, \cdot)$.

Today: a foundational example, then the two main tools for **simplifying** decision problems – sufficiency and convexity.

The Likelihood Principle: a striking example (Berger, Ex. 15)

We are given a coin and are interested in θ, the probability of heads. We want to test $H_0 : \theta = \frac{1}{2}$ vs. $H_1 : \theta > \frac{1}{2}$. An experiment is conducted. The result: 9 heads and 3 tails.

But what was the experiment?

- **Possibility 1:** A fixed $n = 12$ flips. Then $X = [\#\text{ heads}] \sim \mathrm{Bin}(12, \theta)$, and the likelihood is
$$\ell_1(\theta) = \binom{12}{9}\theta^9(1-\theta)^3.$$
- **Possibility 2:** Flip until 3 tails appear. Then $X \sim \mathrm{NegBin}(3, 1-\theta)$, and the likelihood is
$$\ell_2(\theta) = \binom{11}{9}\theta^9(1-\theta)^3.$$

The two likelihoods are **proportional** as functions of θ. The data is the same. But the experiments are very different from a frequentist perspective.

Same data, different p-values

Classical significance testing gives different answers for the two experiments. The p-value is the probability, under $H_0 : \theta = \frac{1}{2}$, of observing a result at least as extreme as what we saw (9 heads).

For the binomial model, "at least as extreme" means $X \geq 9$ out of 12 fixed flips:

$$\alpha_1 = P_{1/2}(X \geq 9) = \sum_{k=9}^{12} \binom{12}{k} (1/2)^{12} = 0.075.$$

For the negative binomial model, "at least as extreme" means needing ≥ 9 heads before seeing 3 tails:

$$\alpha_2 = P_{1/2}(X \geq 9) = \sum_{k=9}^{\infty} \binom{k+2}{k} (1/2)^{k+3} = 0.0325.$$

The p-values differ because the two models assign different probabilities to extreme outcomes, even though they agree on the observed outcome. At the 5% level, the binomial experiment does *not* reject H_0 ($0.075 > 0.05$), while the negative binomial experiment *does* ($0.0325 < 0.05$).

Same coin, same flips, same outcomes, same likelihood – but different conclusions.

The source of the confusion

The classical analysis gave two different answers to the same evidence. Why?

Because the p-value uses information that is **external to the observed data**. It depends on what *could have happened* under the experimental design but did not.

In the binomial model, the experiment could have produced at most 12 heads, so the "extreme" outcomes are $X = 10, 11, 12$. In the negative binomial model, the experiment could have run indefinitely, so the "extreme" outcomes are $X = 10, 11, 12, 13, \ldots$ The p-value sums over these unobserved possibilities, and the two sums give different numbers.

The actual data – 9 heads and 3 tails – is identical. The likelihood functions are proportional. The *only* difference is the set of things that could have happened but did not. The classical analysis is sensitive to this external information. Should it be?

The Likelihood Principle

Definition (The Likelihood Principle)

In making inferences or decisions about θ after x is observed, all relevant experimental information is contained in the likelihood function for the observed x. Two likelihood functions contain the same information about θ if they are proportional to each other (as functions of θ).

There is a principle – the Likelihood Principle – that makes a clean statement: the information about θ in the data lives in the likelihood function $\ell(\theta) = f(x|\theta)$, and nowhere else. Not in the design, not in the sample space, not in what could have happened.

The striking observation from our example is that **classical analysis does not always follow this principle**. The two likelihoods are proportional, so the LP says they carry the same information. But the classical p-values disagree. Classical methods sometimes use more than the likelihood – they use the sampling distribution, which depends on the experimental design.

What this means for us

We are not here to settle the frequentist-Bayesian debate. The frequentist has reasons for caring about the sampling distribution, and those reasons are legitimate in many settings.

But the Likelihood Principle points us toward a concrete question: **what information in the data actually matters for decisions about θ?**

The LP says: the likelihood function. If T is a sufficient statistic, the factorization theorem gives $f(x|\theta) = h(x)\, g(T(x)|\theta)$, so the likelihood depends on x only through $T(x)$.

This means all the information about θ – everything that could affect a good decision – lives in T. That is a claim we can make precise and prove in the decision-theoretic setting. Let us do that now.

Sufficiency: definition and factorization

Definition

Let X be a random variable whose distribution depends on θ. A function $T = T(X)$ is a **sufficient statistic** for θ if the conditional distribution of X given $T(X) = t$ does not depend on θ (with probability one).

Factorization theorem: T is sufficient for θ if and only if

$$f(x|\theta) = h(x)\, g(T(x)|\theta)$$

for some functions h (not depending on θ) and g.

Example: If $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, then $T = (\bar{X}, S^2)$ is sufficient for $\theta = (\mu, \sigma^2)$.

The partition induced by T

If $T(X)$ is a statistic with range $\mathcal{T}$, it partitions the sample space $\mathcal{X}$ into sets

$$\mathcal{X}_t = \{x \in \mathcal{X} : T(x) = t\}, \quad t \in \mathcal{T}.$$

These sets are disjoint and cover $\mathcal{X}$.

A decision rule δ is said to be **based on** T if $\delta(x)$ depends on x only through $T(x)$, i.e., $\delta(x_1) = \delta(x_2)$ whenever $T(x_1) = T(x_2)$.

The question for decision theory is: can we always restrict attention to rules based on a sufficient statistic T, without any loss in risk?

Theorem 1: sufficiency reduces the decision problem (Berger, Thm. 1)

Theorem

Assume that T is a sufficient statistic for θ, and let $\delta_0^*(x, \cdot)$ be any randomized rule in $\mathcal{D}^*$. Then there exists a randomized rule $\delta_1^*(t, \cdot)$, depending only on $T(x)$, which is R-equivalent to δ_0^*.

The equivalent rule based on T is constructed by averaging the original rule over the conditional distribution of X given $T = t$:

$$\delta_1^*(t, A) = E^{X|T=t}[\delta_0^*(X, A)].$$

Sufficiency of T ensures this conditional expectation does not depend on θ, so δ_1^* is a well-defined decision rule.

Proof of Theorem 1: the key step

We need to show $R(\theta, \delta_1^*) = R(\theta, \delta_0^*)$ for all θ.

The loss of δ_1^* at a given t: since $\delta_1^*(t, \cdot)$ is a distribution on $\mathcal{A}$, its loss is the expected loss when a is drawn from it:

$$L(\theta, \delta_1^*(t, \cdot)) = E^{\delta_1^*(t,\cdot)}[L(\theta, a)].$$

Now, recall how δ_1^* was built: $\delta_1^*(t, A) = E^{X|T=t}[\delta_0^*(X, A)]$. This means $\delta_1^*(t, \cdot)$ is a **mixture distribution** on $\mathcal{A}$. It mixes together the distributions $\delta_0^*(x, \cdot)$ for all x with $T(x) = t$, weighted by $P(X = x \mid T = t)$.

Drawing a from this mixture is the same as:

Step 1: Pick which component of the mixture you are in: draw $X \sim (X \mid T = t)$.
Step 2: Draw a from that component: $a \sim \delta_0^*(X, \cdot)$.

Averaging the loss over this two-stage draw gives the iterated expectation:

$$E^{\delta_1^*(t,\cdot)}[L(\theta, a)] = E^{X|T=t}\left[E^{\delta_0^*(X,\cdot)}[L(\theta, a)]\right].$$

The rest follows by taking expectations.

Proof of Theorem 1: completing the argument

We have established that $L(\theta, \delta_1^*(t, \cdot)) = E^{X|T=t}\left[E^{\delta_0^*(X,\cdot)}[L(\theta, a)]\right]$.

Now take the risk by averaging over T:

$$\begin{aligned} R(\theta, \delta_1^*) &= E^T[L(\theta, \delta_1^*(T, \cdot))] \\ &= E^T\left[E^{X|T}\left[E^{\delta_0^*(X,\cdot)}[L(\theta, a)]\right]\right] \\ &= E^X\left[E^{\delta_0^*(X,\cdot)}[L(\theta, a)]\right] \quad \text{(tower property)} \\ &= E^X[L(\theta, \delta_0^*(X, \cdot))] = R(\theta, \delta_0^*). \quad \square \end{aligned}$$

The tower property collapses the iterated expectation $E^T[E^{X|T}[\cdot]]$ into $E^X[\cdot]$. After that, we just recognize the definition of $R(\theta, \delta_0^*)$.

Example: Theorem 1 in action

Let $X_1, X_2 \overset{\text{iid}}{\sim} \mathrm{Bernoulli}(\theta)$. The sufficient statistic is $T = X_1 + X_2$.

Consider the rule $\delta_0(x_1, x_2) = x_1$ (just use the first observation to estimate θ). This is *not* a function of T: when $T = 1$, the data could be $(1, 0)$ or $(0, 1)$, and δ_0 gives different values.

Construct the equivalent rule based on T:
$\delta_1^*(t, A) = P(\delta_0(X) \in A \mid T = t) = P(X_1 \in A \mid T = t)$.

Since δ_0 is nonrandomized, this asks: given that $T = t$, what is the distribution of the action $\delta_0(X) = X_1$? We need the conditional distribution of X_1 given $X_1 + X_2 = t$.

Example continued: computing δ_1^* case by case

The sample space is $\{(0,0), (0,1), (1,0), (1,1)\}$.

$T = 0$**:** The only possibility is $(X_1, X_2) = (0,0)$. So $X_1 = 0$ with certainty, and δ_1^* takes action 0 with probability 1.

$T = 2$**:** The only possibility is $(X_1, X_2) = (1,1)$. So $X_1 = 1$ with certainty, and δ_1^* takes action 1 with probability 1.

$T = 1$**:** Two possibilities: $(1,0)$ and $(0,1)$. By sufficiency, the conditional distribution of (X_1, X_2) given $T = 1$ does not depend on θ. Since X_1 and X_2 are exchangeable,

$$P(X_1 = 1 \mid T = 1) = P(X_1 = 0 \mid T = 1) = \tfrac{1}{2}.$$

So δ_1^* takes action 1 with probability $\frac{1}{2}$ and action 0 with probability $\frac{1}{2}$.

The original rule δ_0 was nonrandomized, but the equivalent rule δ_1^* based on T is **genuinely randomized** when $T = 1$: it flips a fair coin between the two actions.

Example continued: verifying R-equivalence

Loss: $L(\theta, a) = (\theta - a)^2$. Compute both risks.

Original rule $\delta_0(X_1, X_2) = X_1$:

$$R(\theta, \delta_0) = E_{P_\theta}[(\theta - X_1)^2] = \mathrm{Var}(X_1) = \theta(1-\theta).$$

New rule δ_1^* based on T: the risk is $R(\theta, \delta_1^*) = E^T[L(\theta, \delta_1^*(T, \cdot))]$. Since $T \in \{0, 1, 2\}$, this is a sum over three terms, each of the form $P_\theta(T = t) \cdot L(\theta, \delta_1^*(t, \cdot))$:

t	$P_\theta(T = t)$	action distribution	$L(\theta, \delta_1^*(t, \cdot))$
0	$(1-\theta)^2$	$a = 0$ w.p. 1	$(\theta - 0)^2 = \theta^2$
1	$2\theta(1-\theta)$	$a = 0$ or 1, each w.p. $\frac{1}{2}$	$\frac{1}{2}\theta^2 + \frac{1}{2}(1-\theta)^2$
2	θ^2	$a = 1$ w.p. 1	$(\theta - 1)^2 = (1-\theta)^2$

$$R(\theta, \delta_1^*) = (1-\theta)^2 \cdot \theta^2 + 2\theta(1-\theta) \cdot \left[\tfrac{1}{2}\theta^2 + \tfrac{1}{2}(1-\theta)^2\right] + \theta^2 \cdot (1-\theta)^2 = \theta(1-\theta).$$

R-equivalent, as Theorem 1 guarantees.

Where we stand with this example

We started with $\delta_0(x_1, x_2) = x_1$. Theorem 1 gave us δ_1^* based on T with the same risk $\theta(1-\theta)$.

But δ_1^* is randomized when $T = 1$. We traded a clean nonrandomized rule for a randomized one with the same risk. That feels unsatisfying.

Can we do better? We will return to this example once we have the right tool.

What Theorem 1 means

When evaluating decision rules through risk functions, **it is only necessary to consider rules based on a sufficient statistic**.

If a rule is not a function of the sufficient statistic, we can always find another rule that is a function of the sufficient statistic and has the same risk function. In other words:

> *The class of (randomized) decision rules based on a sufficient statistic is an* **essentially complete class**.

Instead of searching over all functions $\delta : \mathcal{X} \to \mathcal{A}$, we can restrict to functions $\delta : \mathcal{T} \to \mathcal{A}$, where $\mathcal{T}$ is the range of T. When T reduces the dimension of the data substantially (as it often does), this makes the problem much more tractable.

A caution

Theorem 1 says we can find an R-*equivalent* rule based on T. It does not say the original rule is inadmissible.

However, it will often be the case that a decision rule which is *not* solely a function of the sufficient statistic is in fact inadmissible. We will see concrete instances of this shortly, once we have the convexity tools.

It is also worth noting that Theorem 1 holds for randomized rules. The equivalent rule δ_1^* based on T may itself be randomized, even if the original rule δ_0 was not. This is because the averaging over $\mathcal{X}_t$ can spread probability across actions.

Convex sets and convex functions

Definition

A set $\Omega \subset \mathbb{R}^m$ is **convex** if for any $x, y \in \Omega$ and $0 \leq \alpha \leq 1$,

$$\alpha x + (1 - \alpha) y \in \Omega.$$

Definition

A real-valued function g defined on a convex set Ω is **convex** if

$$g(\alpha x + (1 - \alpha) y) \leq \alpha\, g(x) + (1 - \alpha)\, g(y)$$

for all $x, y \in \Omega$ and $0 < \alpha < 1$. It is **strictly convex** if the inequality is strict whenever $x \neq y$.

The **convex hull** of a set $S \subset \mathbb{R}^m$ is the set of all convex combinations $\sum_i \alpha_i x_i$ with $\alpha_i \geq 0$, $\sum_i \alpha_i = 1$, and $x_i \in S$. It is the smallest convex set containing S.

Jensen's inequality (Berger, Thm. 2)

Theorem (Jensen's Inequality)

Let $g : \Omega \to \mathbb{R}$ be a convex function on a convex set $\Omega \subset \mathbb{R}^m$, and let X be a random vector with $E[|X|] < \infty$ and $P(X \in \Omega) = 1$. Then

$$g(E[X]) \leq E[g(X)],$$

with strict inequality if g is strictly convex and X is not concentrated at a point.

Example: Since $g(x) = x^2$ is strictly convex on $\mathbb{R}$, Jensen gives $(E[X])^2 < E[X^2]$ whenever X has finite mean and is not a constant. This is exactly the statement $\mathrm{Var}(X) > 0$.

Why Jensen matters for decision theory

Jensen's inequality will be used in two ways throughout the course:

1. Eliminating randomization. If the loss function $L(\theta, a)$ is convex in a, we can show that randomized decision rules are unnecessary: for any randomized rule, there exists a nonrandomized rule with loss that is *pointwise* no larger.

2. The Rao–Blackwell theorem. If the loss is convex and a sufficient statistic T exists, we can improve any decision rule by conditioning on T. The improved rule is nonrandomized, based on T, and has risk no larger than the original.

We develop these two consequences now.

Theorem 3: convex loss makes randomization unnecessary (Berger, Thm. 3)

Theorem

Assume $\mathcal{A}$ is a convex subset of $\mathbb{R}^m$, and that for each $\theta \in \Theta$, the loss $L(\theta, a)$ is a convex function of a. Let δ^* be a randomized decision rule with $E^{\delta^*(x,\cdot)}[|a|] < \infty$ for all x. Then the nonrandomized rule

$$\delta(x) = E^{\delta^*(x,\cdot)}[a]$$

satisfies $L(\theta, \delta(x)) \leq L(\theta, \delta^*(x, \cdot))$ for all x and θ.

In particular, $R(\theta, \delta) \leq R(\theta, \delta^*)$ for all θ.

Proof of Theorem 3

The proof is a direct application of Jensen's inequality.

Since $\mathcal{A}$ is convex and $\delta^*(x, \cdot)$ is a probability distribution on $\mathcal{A}$, we have $\delta(x) = E^{\delta^*(x,\cdot)}[a] \in \mathcal{A}$ (by the convexity lemma: the mean of a random vector supported on a convex set lies in that set).

Then, for each θ and x:

$$L(\theta, \delta(x)) = L\Big(\theta,\ E^{\delta^*(x,\cdot)}[a]\Big) \leq E^{\delta^*(x,\cdot)}[L(\theta, a)] = L(\theta, \delta^*(x, \cdot)).$$

The inequality is Jensen, applied to the convex function $a \mapsto L(\theta, a)$. $\square$

If L is *strictly* convex, the inequality is strict whenever $\delta^*(x, \cdot)$ is nondegenerate. So the nonrandomized rule is *strictly better* at every x where actual randomization occurs.

Returning to our example: derandomizing δ_1^*

Recall: $X_1, X_2 \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$, $T = X_1 + X_2$, squared-error loss $L(\theta, a) = (\theta - a)^2$.

Theorem 1 gave us δ_1^* based on T, which is R-equivalent to $\delta_0(x_1, x_2) = x_1$ but randomized when $T = 1$: it takes action 0 or 1 each with probability $\frac{1}{2}$.

Squared-error loss is strictly convex in a. Apply Theorem 3. The derandomized rule is:

$$\delta(t) = E^{\delta_1^*(t,\cdot)}[a].$$

- $T = 0$: δ_1^* puts all mass on 0. So $\delta(0) = 0$.
- $T = 2$: δ_1^* puts all mass on 1. So $\delta(2) = 1$.
- $T = 1$: δ_1^* puts mass $\frac{1}{2}$ on 0 and $\frac{1}{2}$ on 1. So $\delta(1) = \frac{1}{2}(0) + \frac{1}{2}(1) = \frac{1}{2}$.

The derandomized rule is $\delta(t) = t/2 = \bar{X}$: the sample mean.

The improvement is strict

Compare the risks:

Original rule $\delta_0(X_1, X_2) = X_1$ and the R-equivalent δ_1^*:
$R(\theta, \delta_0) = R(\theta, \delta_1^*) = \theta(1-\theta)$.

Derandomized rule $\delta(T) = T/2 = \bar{X}$:

$$R(\theta, \delta) = E_{P_\theta}[(\theta - \bar{X})^2] = \mathrm{Var}(\bar{X}) = \frac{\theta(1-\theta)}{2}.$$

$R(\theta, \delta) = \frac{1}{2} R(\theta, \delta_0)$ for all θ. Strictly smaller (whenever $\theta \neq 0, 1$).

What happened? Theorem 1 gave us an R-equivalent rule on T, but it was randomized. Theorem 3 removed the randomization, and the strict convexity of squared-error loss turned R-equivalence into a strict improvement.

There is a way to do both steps at once. That is the Rao–Blackwell theorem.

Another example: a mixture of two estimators (Berger, Ex. 20)

Let $X \sim \mathrm{Bin}(n, \theta)$, $\mathcal{A} = \mathbb{R}$, and $L(\theta, a) = (\theta - a)^2$ (strictly convex in a).

Two estimators: $\delta_1(x) = x/n$ and $\delta_2(x) = 1$.

The **randomized rule** δ^*: for each observed x, flip a fair coin. If heads, take action $\delta_1(x) = x/n$. If tails, take action $\delta_2(x) = 1$. In the notation of Definition 7:

$$\delta^*(x, \{a\}) = \begin{cases} \frac{1}{2} & \text{if } a = x/n \\ \frac{1}{2} & \text{if } a = 1 \end{cases} \qquad (\text{when } x/n \neq 1).$$

The **nonrandomized (derandomized) rule** from Theorem 3:

$$\delta(x) = E^{\delta^*(x,\cdot)}[a] = \tfrac{1}{2}(x/n) + \tfrac{1}{2}(1) = \frac{x/n + 1}{2}.$$

This is a deterministic action for each x: always take the midpoint.

Comparing the losses

Fix θ and x. Write $\hat{\theta} = x/n$ for brevity.

Randomized rule: the loss averages over the coin flip:

$$L(\theta, \delta^*(x, \cdot)) = \tfrac{1}{2}(\theta - \hat{\theta})^2 + \tfrac{1}{2}(\theta - 1)^2.$$

Derandomized rule: the loss is deterministic:

$$L(\theta, \delta(x)) = \left(\theta - \frac{\hat{\theta} + 1}{2}\right)^2.$$

Let $u = \theta - \hat{\theta}$ and $v = \theta - 1$. Then the randomized loss is $\frac{1}{2}u^2 + \frac{1}{2}v^2$, the derandomized loss is $\left(\frac{u+v}{2}\right)^2$, and the difference is:

$$\tfrac{1}{2}u^2 + \tfrac{1}{2}v^2 - \frac{(u+v)^2}{4} = \frac{u^2 - 2uv + v^2}{4} = \frac{(u-v)^2}{4} = \frac{(1-\hat{\theta})^2}{4} \geq 0.$$

This does not depend on θ or n. It is strictly positive whenever $\hat{\theta} \neq 1$, i.e., whenever the randomization is nondegenerate. That is Jensen's inequality at work.

Combining sufficiency and convexity

We now have two tools:

- **Theorem 1** (sufficiency): any rule can be replaced by an R-equivalent rule based on T.
- **Theorem 3** (convexity): under convex loss, any randomized rule can be replaced by a nonrandomized rule with no larger loss.

We saw this in the Bernoulli example: $\delta_0 \to \delta_1^*$ (R-equivalent, randomized, based on T) $\to \delta = \bar{X}$ (nonrandomized, based on T, strictly better).

Let us now trace the two-step pipeline in general and see what formula comes out.

The pipeline produces a specific formula

Start with any nonrandomized rule $\delta_0(x)$.

Step 1 (Theorem 1): Build the R-equivalent randomized rule based on T:

$$\delta_1^*(t, A) = P(\delta_0(X) \in A \mid T = t).$$

This is a distribution on $\mathcal{A}$ for each t: draw X from $X|T = t$, apply δ_0, and see where the action lands.

Step 2 (Theorem 3): Derandomize by taking the expected action:

$$\delta(t) = E^{\delta_1^*(t,\cdot)}[a].$$

But the distribution $\delta_1^*(t, \cdot)$ was built by drawing $X|T = t$ and applying δ_0. So the expected action is:

$$E^{\delta_1^*(t,\cdot)}[a] = E^{X|T=t}[\delta_0(X)].$$

The pipeline mechanically produces $\delta(t) = E[\delta_0(X) \mid T = t]$. The formula is not a new idea. It is what Theorem 1 followed by Theorem 3 gives you.

Formalizing the combined result

The two-step pipeline says: for any nonrandomized rule δ_0 and convex loss, the rule

$$\delta_1(t) = E[\delta_0(X) \mid T = t]$$

is nonrandomized, based on T, and has risk no larger than δ_0.

The risk improvement comes from Step 2 (Jensen/Theorem 3), not from Step 1 (which only gave R-equivalence).

This is exactly the Rao–Blackwell theorem. It packages the pipeline into a single statement.

The Rao–Blackwell theorem (Berger, Thm. 4)

Theorem (Rao–Blackwell)

Assume $\mathcal{A}$ is a convex subset of $\mathbb{R}^m$ and that $L(\theta, a)$ is a convex function of a for all θ. Suppose T is a sufficient statistic for θ, and $\delta_0(x)$ is a nonrandomized decision rule. Then the rule

$$\delta_1(t) = E^{X|T=t}[\delta_0(X)]$$

is R-equivalent to or R-better than δ_0, provided the expectation exists.

In words: condition your estimator on the sufficient statistic. The result is at least as good, and usually strictly better.

Proof of the Rao–Blackwell theorem

By the definition of a sufficient statistic, the conditional expectation $E^{X|T=t}[\delta_0(X)]$ does not depend on θ. So δ_1 is a well-defined decision rule based on T.

By Jensen's inequality (applied to the convex function $a \mapsto L(\theta, a)$):

$$L(\theta, \delta_1(T)) = L\Big(\theta,\ E^{X|T}[\delta_0(X)]\Big) \leq E^{X|T}[L(\theta, \delta_0(X))].$$

Taking expectations over T:

$$\begin{aligned} R(\theta, \delta_1) &= E^T[L(\theta, \delta_1(T))] \\ &\leq E^T\Big[E^{X|T}[L(\theta, \delta_0(X))]\Big] \\ &= E^X[L(\theta, \delta_0(X))] \quad \text{(tower property)} \\ &= R(\theta, \delta_0). \quad \square \end{aligned}$$

What the Rao–Blackwell theorem gives us

The theorem has two readings:

As a tool for improving estimators. Given any estimator δ_0, we can improve it (in terms of risk) by conditioning on the sufficient statistic. The improved estimator $\delta_1(t) = E[\delta_0(X)|T = t]$ is always at least as good, and usually strictly better.

As a structural result. When the loss is convex and a sufficient statistic T exists, we can restrict attention to nonrandomized rules based on T without any loss in risk. The search space for good decision rules is dramatically reduced.

Combining Theorems 1 and 3: *under convex loss with a sufficient statistic T, only nonrandomized rules based on T need be considered.*

Completing our running example via Rao–Blackwell

$X_1, X_2 \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$, $T = X_1 + X_2$ sufficient, $L(\theta, a) = (\theta - a)^2$.

We started with $\delta_0(x_1, x_2) = x_1$. Apply Rao–Blackwell directly:

$$\delta_1(t) = E[X_1 \mid T = t] = E[X_1 \mid X_1 + X_2 = t].$$

By symmetry, $E[X_1 \mid X_1 + X_2 = t] = E[X_2 \mid X_1 + X_2 = t] = t/2$.

So $\delta_1(t) = t/2 = \bar{X}$, exactly the same answer we obtained by doing Theorem 1 then Theorem 3 in sequence. Rao–Blackwell does both steps at once.

The full trajectory of this example:

$\delta_0(x_1, x_2) = x_1$	not based on T	$R(\theta, \delta_0) = \theta(1-\theta)$
$\delta_1^*(t, \cdot)$ via Thm 1	based on T, randomized	$R(\theta, \delta_1^*) = \theta(1-\theta)$
$\delta(t) = t/2$ via Thm 3	based on T, nonrandomized	$R(\theta, \delta) = \theta(1-\theta)/2$
$\delta_1(t) = t/2$ via R-B	based on T, nonrandomized	$R(\theta, \delta_1) = \theta(1-\theta)/2$

Sufficiency gave R-equivalence. Convexity gave strict improvement.

Another example: improving the sample median

Let $X = (X_1, \ldots, X_n)$ with $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\theta, 1)$. Estimate θ under squared-error loss.

The sufficient statistic is $T = \bar{X}$.

Consider $\delta_0(X) = \widetilde{X}$, the sample median. This is not a function of $\bar{X}$ alone (it depends on the order statistics, not just the mean). The Rao–Blackwell theorem says we can improve it:

$$\delta_1(\bar{x}) = E[\widetilde{X} \mid \bar{X} = \bar{x}].$$

A symmetry argument (the conditional distribution of $(X_1 - \bar{X}, \ldots, X_n - \bar{X})$ given $\bar{X}$ is symmetric) shows that $E[\widetilde{X} \mid \bar{X}] = \bar{X}$.

The Rao–Blackwellized estimator is the sample mean. Since squared-error loss is strictly convex, the sample mean has *strictly smaller* risk than the sample median for all θ. (This is Berger's Exercise 20c.)

The big picture from today

Three results from today:

1. **Sufficiency (Theorem 1):** Any rule can be replaced by an R-equivalent rule based on the sufficient statistic T. The class of rules based on T is essentially complete.

2. **Convex loss eliminates randomization (Theorem 3):** Under convex loss, any randomized rule can be replaced by a nonrandomized rule with no greater risk.

3. **Rao–Blackwell (Theorem 4):** Under convex loss with a sufficient statistic, condition on T to improve any estimator.

Combined: under convex loss, only nonrandomized decision rules based on the sufficient statistic need be considered. This is the starting point for all the optimization that follows.